

Yuri Shiroiva

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with 2+ years across the full stack of applied AI: LLM multi-agent orchestration, large-scale time-series pipelines, and deep learning research. Ship LLM-powered diagnostics on Google Cloud Vertex AI over a Cassandra telemetry platform, and design agentic systems with the Model Context Protocol (MCP) and schema-constrained structured output. Specialized in industrial protocol reverse engineering (CAN bus / ISOBUS), a scarce differentiator for vehicle, IoT, and machine-telemetry domains.

CORE SKILLS

AI & LLM Systems: Multi-agent orchestration, Anthropic Claude API via Vertex AI, Model Context Protocol (MCP) tool servers, structured output with JSON Schema, prompt engineering, context/token optimization (TOON, YAML), LLM-driven diagnostics, RAG pipelines, LangChain, Hugging Face Transformers, transformer fine-tuning

Machine Learning & Deep Learning: PyTorch, scikit-learn, LightGBM, CNNs, semantic segmentation (U-Net, DeepLabV3+, ResNet encoders), multi-task learning, uncertainty weighting, test-time augmentation, model ensembling, OOF cross-validation

Data Engineering: Python, SQL, pandas, NumPy, ETL/ELT pipeline design, high-volume time-series ingestion, Apache Cassandra (bi-temporal modeling, TIMEUUID clustering), PostgreSQL, data validation and quality routines, geospatial data (GeoJSON, shapefiles, Sentinel-2 raster stacks)

Cloud & Platform: Google Cloud Platform (Vertex AI, VPC, IAP tunneling, service-account IAM), AWS, AWS-to-GCP migration, Docker, REST API design, backend services

Domain Expertise: CAN bus and ISOBUS protocol engineering (J1939 / ISO 11783), proprietary PGN reverse engineering, agricultural machinery telemetry, precision agriculture, task controller integration

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer, IoT (AgTech IoT platform) 2024 - Present

Agricultural machinery telemetry platform. Own the data pipeline and applied-AI layer end to end.

- Designed and shipped an LLM-powered machine diagnostics feature end to end, taking Cassandra telemetry through context summarization into the Claude API on Vertex AI and returning schema-validated structured output in the product UI.
- Architected a multi-agent LLM system for reverse-engineering undocumented CAN protocol messages, orchestrating parallel compute kernels through MCP tool servers with ground-truth validation anchors and hold-out testing to prevent overfitting.
- Built and maintained Python pipelines that transform raw CAN/ISOBUS telemetry into structured datasets, geospatial coverage maps (GeoJSON), and standards-compliant ISOXML task files consumed by downstream operations.
- Designed an append-only bi-temporal versioning schema in Cassandra (TIMEUUID clustering with validity intervals) giving full configuration traceability for every field operation, replacing a destructive-update model that lost audit history.
- Reverse-engineered an encrypted third-party telemetry format (AES-128-CBC) to unlock vendor-locked geospatial data, expanding platform coverage without requiring partner API access.
- Diagnosed and fixed production defects in the ingestion service, including a unit-conversion error in flow-delay compensation and an aggregation-scoping bug in applied-product totals that corrupted downstream operational reports.
- Implemented a decorator-based plugin registry enabling multi-vendor equipment support (John Deere, Massey Ferguson, Fendt, Horsch, Kuhn, Trimble) without modifying core pipeline code; supported the AWS-to-GCP migration and set up secure IAP-tunneled access to VPC-isolated PostgreSQL.

Software Development Intern, Optcore 2023 - 2024

- Developed and maintained backend services and REST APIs powering core web application features.
- Designed cleaning, transformation, and validation routines for data ingested from heterogeneous sources.
- Applied Levenshtein distance and fuzzy-matching techniques to deduplicate records and raise data quality.

RESEARCH

Undergraduate Thesis, PUC-PR: Semantic Segmentation of Agricultural Field Boundaries 2025 - 2026

- Built a deep learning ensemble (U-Net + DeepLabV3+ with ResNet encoders) for delineating field boundaries from 30-channel multitemporal Sentinel-2 satellite imagery, trained on the AI4Boundaries dataset.
- Implemented multi-task learning with three prediction heads (extent, boundary, distance) using learned heteroscedastic uncertainty weighting to balance losses automatically instead of hand-tuned coefficients.
- Engineered a post-processing stack (test-time augmentation, empty-field gating, gradient-boosted false-positive suppressor) and evaluated with bootstrap 95% confidence intervals, Cohen's d effect sizes, and area-bias analysis.

EDUCATION

Technology Degree in Applied Artificial Intelligence, Pontifical Catholic University of Paraná (PUC-PR) 2024 - 2026
Weighted GPA: 9.27 / 10. **Honorable Mention (Menção Honrosa)** for overall academic performance, PUC-PR

Electrical Engineering (Robotics & Industrial Automation), Federal University of Juiz de Fora (UFJF) 2021 - 2023
Incomplete. Transferred to Applied Artificial Intelligence at PUC-PR in 2024.

CERTIFICATIONS & LANGUAGES

Professional Certificates: IBM Generative AI Engineering (2026) · IBM Data Science (2024) · Google Data Analytics (2023)

Portuguese: Native | English: Basic conversational, proficient technical reading | Spanish: Basic